

2)

A) CODE OUTPUT:

```
nithya@WIN-482M0QMA856:~$ cc practical2.c
nithya@WIN-482M0QMA856:~$ ./a.out
Files successfully opened
File copied successfully
nithya@WIN-482M0QMA856:~$
```

Contents of source.txt

```
nithya@WIN-482M0QMA856:~$ cat source.txt
hello
this is os
practical
```

Contents of destination.txt

```
nithya@WIN-482M0QMA856:~$ cat destination.txt
hello
this is os
practical
```

B) Commands

```
nithya@WIN-482M0QMA856:~$ strace cat sample.txt
execve("/usr/bin/cat", ["cat", "sample.txt"], 0x7ffdbd7bc4d8 /* 25 vars */) = 0
brk(NULL)                                = 0x57f2a9be9000
mmap(NULL, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7eb370055000
access("/etc/ld.so.preload", R_OK)       = -1 ENOENT (No such file or directory)
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=27889, ...}) = 0
mmap(NULL, 27889, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7eb37004e000
close(3)                                 = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0\0\0\0...", 832) = 832
fstat(3, {st_mode=S_IFREG|0644, st_size=199120, ...}) = 0
mmap(NULL, 206288, PROT_READ, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7eb37001b000
mmap(0x7eb370022000, 135168, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x7000) = 0x7eb370022000
mmap(0x7eb370043000, 28672, PROT_READ, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x28000) = 0x7eb370043000
mmap(0x7eb37004a000, 8192, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x2f000) = 0x7eb37004a000
mmap(0x7eb37004c000, 5584, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS,
```

```

hithya@WIN-482M0QMA856:~$ uname
Linux
hithya@WIN-482M0QMA856:~$ lscpu
Architecture:            x86_64
CPU op-mode(s):          32-bit, 64-bit
Address sizes:            48 bits physical, 48 bits virtual
Byte Order:               Little Endian
CPU(s):                   16
On-line CPU(s) list:      0-15
Vendor ID:                 AuthenticAMD
Model name:                AMD Ryzen 7 7730U with Radeon Graphics
CPU family:                25
Model:                     80
Thread(s) per core:        2
Core(s) per socket:        8
Socket(s):                 1
Stepping:                  0
BogoMIPS:                  3992.39
Flags:                     fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge m
ca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall
nx mmxext fxsr_opt pdpe1gb rdtscp lm constant_tsc rep
_good nopl xtopology tsc_reliable nonstop_tsc cpuid ex
td_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16
sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand h
ypervisor lahf_lm cmp_legacy svm cr8_legacy abm sse4a
misalignsse 3dnowprefetch osvw topoext perfctr_core ss
bd ibrs ibpb stibp vmmcall fsgsbase bmi1 avx2 smep bmi
2 erms invpcid rdseed adx smap clflushopt clwb sha_ni
xsaveopt xsavec xgetbv1 xsaves clzero xsaveerptr arat
npt nrip_save tsc_scale vmcb_clean flushbyasid decodea
ssists pausefilter pfthreshold v_vmsave_vmload umip va
es vpcmlulqdq rdpid fsrm
Virtualization features:
Virtualization:            AMD-V
Hypervisor vendor:         Microsoft
Virtualization type:        full
Caches (sum of all):
L1d:                        256 KiB (8 instances)
L1i:                        256 KiB (8 instances)
L2:                          4 MiB (8 instances)
L3:                          16 MiB (1 instance)
NUMA:
NUMA node(s):               1
NUMA node0 CPU(s):          0-15

```

```

nithya@WIN-482M0QMA856:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda   8:0    0 356.9M  1 disk
sdb   8:16   0 159.4M  1 disk
sdc   8:32   0    2G    0 disk [SWAP]
sdd   8:48   0    1T    0 disk /mnt/wslg/distro
/

nithya@WIN-482M0QMA856:~$ ps
  PID TTY          TIME CMD
  364 pts/0    00:00:00 bash
 1980 pts/0    00:00:00 ps

nithya@WIN-482M0QMA856:~$ top
top - 13:33:22 up 59 min,  1 user,  load average: 0.00, 0.00, 0.00
Tasks:  25 total,   1 running, 24 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  6849.6 total,  6095.2 free,   609.1 used,   299.6 buff/cache
MiB Swap:  2048.0 total,  2048.0 free,    0.0 used,  6240.5 avail Mem

   PID USER      PR  NI   VIRT   RES   SHR  S  %CPU  %MEM     TIME+ COMMAND
    1 root        20   0   24284   15372 11516 S   0.0   0.2   0:01.89 systemd
    2 root        20   0    3180    2212   2072 S   0.0   0.0   0:00.02 init-syste+
    6 root        20   0    3180    2040   1940 S   0.0   0.0   0:00.00 init
   52 root        19  -1   50400   16764 15464 S   0.0   0.2   0:00.54 systemd-jo+
   69 systemd+    20   0   22416   14444 11944 S   0.0   0.2   0:00.18 systemd-re+
   91 root        20   0   35260   12288   9176 S   0.0   0.2   0:03.17 systemd-ud+

```

Command

Role/Purpose

uname	Displays the os being used.
lscpu	Displays CPU architecture, cores, threads, and processor information.
lsblk	Displays storage devices, partitions, and their sizes.
ps	Displays currently running processes and their PIDs.
top	Provides real-time information about processes and CPU/memory usage.